Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details

Batch Number 11

Start date 29-DEC-25

Names:

Internship Duration: 8 Weeks

1. Project Title

FOOD TRENDS UNDERSTANDING CUSTOMER PREFERENCES IN FB

2. Project Objective

- The primary objective of the project is to analyze customer feedback and behavioral data to uncover meaningful insights about food preferences, satisfaction levels, and emerging consumption trends.
- The project aimed to integrate structured and unstructured feedback data into interactive dashboards developed using Microsoft Power BI.
- By leveraging sentiment analysis and data visualization techniques, the project sought to support marketing, product development, and strategy teams in making informed, data-driven decisions.
- Understanding customer preferences in the food and beverage (FB) sector is critical for enhancing product quality, improving customer retention, and identifying innovation opportunities.

3. Project description in detail

This project focused on building a centralized analytics system to understand customer preferences in the FB sector. Data was collected from multiple sources, including survey responses, product reviews, and social media feedback. These datasets included structured attributes such as ratings, regions, and purchase frequency, along with unstructured textual comments.

The initial phase involved cleaning and integrating the datasets using Power Query. Duplicate records were removed, missing values were handled, and inconsistent column formats were standardized. Text normalization techniques were applied to prepare feedback for sentiment analysis. A unified data model was created with key entities such as Customer, Product, Feedback, and Time dimensions, ensuring smooth relational mapping and cross-filtering within the dashboard.

Sentiment analysis was implemented using Python-based natural language processing methods to classify feedback as positive, neutral, or negative. Sentiment scores were then integrated into Power BI, where DAX measures were developed to compute overall sentiment ratios, product-wise satisfaction scores, and sentiment trends over time.

The dashboard consisted of multiple analytical layers. The sentiment overview page presented overall satisfaction metrics and regional breakdowns. The product insights module compared features such as taste, price, packaging, and quality across different food categories. A customer segmentation module grouped users into Loyal Customers, Critics, and Occasional Buyers based on their feedback frequency and sentiment patterns. The final strategy layer highlighted trending keywords, recurring complaints, and shifts in preferences before and after campaigns, enabling

data-driven recommendations for business growth.

3. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Requirement gathering & Data Collection	Collected Survey and review datasets
Week 2	Data cleaning & schema designs	Created unified data model
Week 3	Sentiment model development	Implement NLP Clasification
Week 4	Sentiment dashboard creation	Built sentiment trend visuals.
Week 5	Product insights module	Feature-based comparision dashboards
Week 6	Correlation and performance analysis	Sales vs Sentiment dashboards
Week 7	Customer segmentation logic	Behavioral personas created
Week 8	Final strategy and dashboard presentation	Final dashboard and documentation completed

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Data sources from surveys, social media, and reviews are cleaned, de-duplicated, and standardized. Unified data model with consistent keys (e.g., product_id, user_id) is validated for future transformations.	**10-JAN-2026**
Prototype/First Draft	Sentiment analysis logic is implemented and verified. Dashboard includes overall sentiment trends and keyword-based feedback themes.	**30-JAN-2026**
Mid-Term Review	Product-level performance dashboards are implemented with category-level comparisons(e.g., taste, price, packaging). Cross-filtering and correlation with sentiment are functional.	**05-FEB-2026**
Final Submission	Segmentation logic based on sentiment behavior and frequency is operational. User personas are visualized across filters like region and age group.	**25-FEB-2026**
Presentation	Final strategy dashboard includes emerging trend keywords, sentiment shifts before/after campaigns, and data-driven recommendations. Project documentation and stakeholder-ready presentation are complete.	**05-MAR-2026**

5b. Project execution details

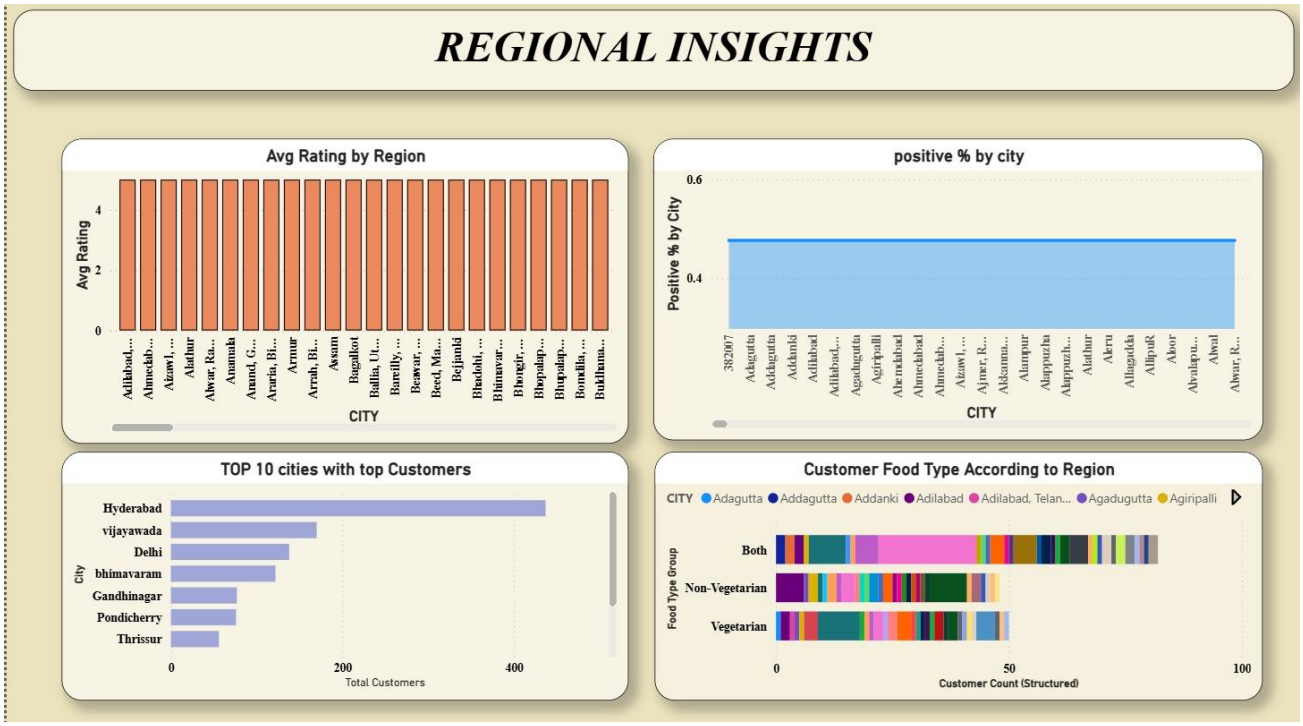
The project execution followed a systematic and milestone-based approach. The first phase involved requirement analysis and identification of key performance indicators relevant to understanding customer preferences. Once the objectives were defined, datasets were gathered and processed to ensure consistency and reliability.

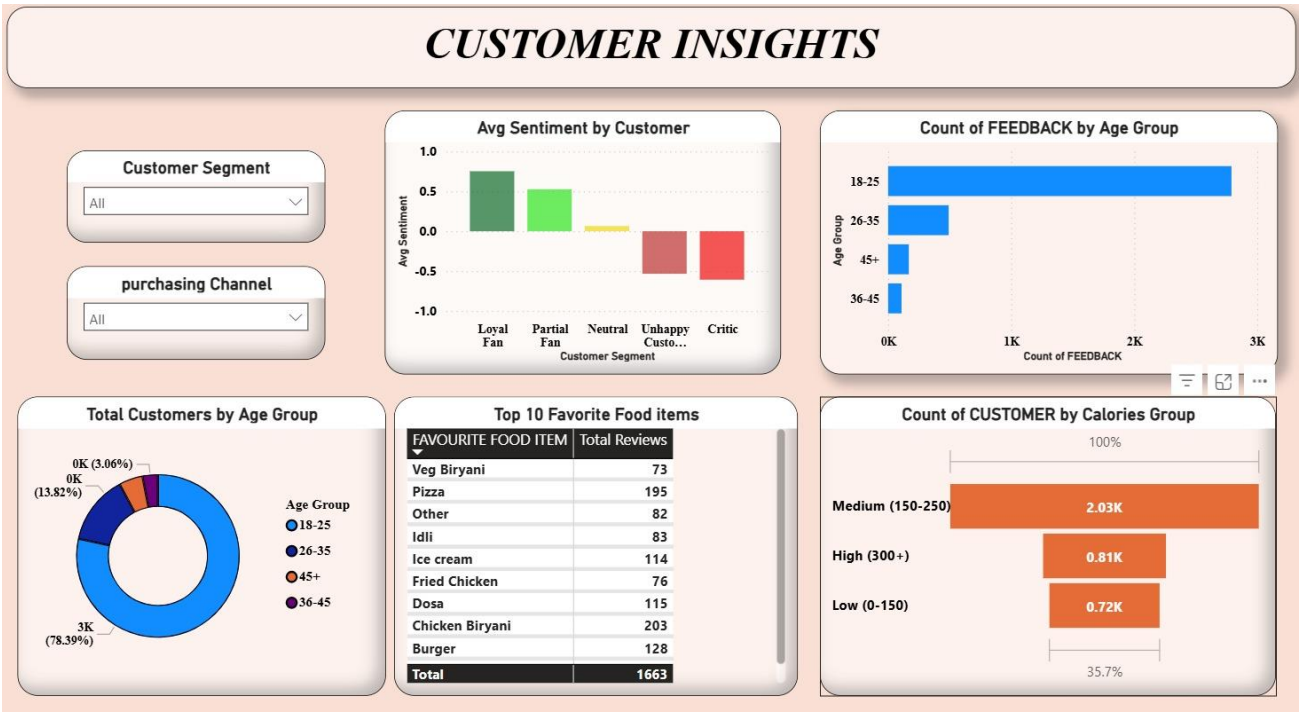
During the development phase, sentiment analysis scripts were created and tested to classify customer feedback accurately. The processed data was then imported into Power BI, where relationships between tables were established to form a semantic data model. DAX calculations were implemented to derive performance metrics and trend indicators.

Dashboard design focused on clarity, interactivity, and usability. Filters and slicers were incorporated to allow dynamic exploration by region, product category, and time period. Continuous validation was performed to ensure data accuracy and logical consistency across visualizations. Performance optimization techniques were applied to enhance dashboard responsiveness when working with large datasets.

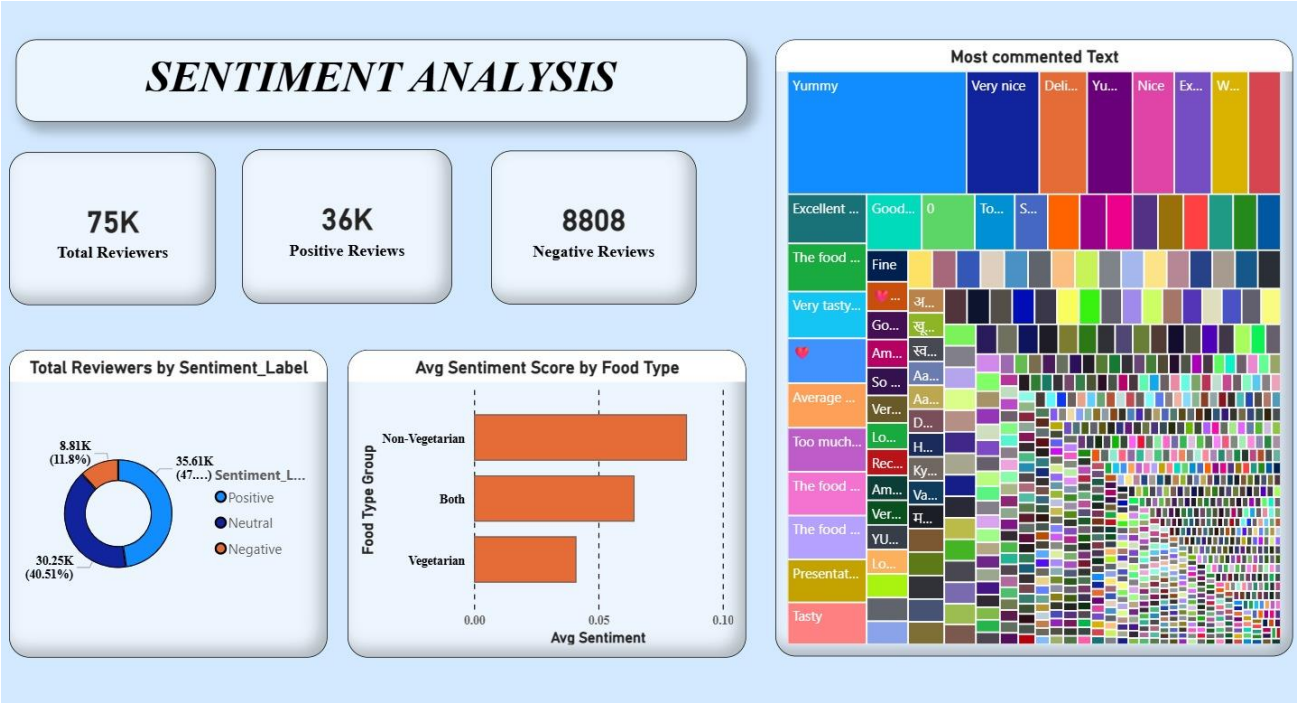
The final stage involved consolidating all modules into a single integrated dashboard and preparing documentation that explained the architecture, methodology, and insights derived from the analysis.

6.Snapshots / Screenshots


REGIONAL INSIGHTS
Avg Rating by Region
Avg Rating
CITY
positive % by city
Positive % by City
CITY
TOP 10 cities with top Customers
Hyderabad
vijayawada
Delhi
bhimavaram
Gandhinagar
Pondicherry
Thrissur
City
Total Customers
Customer Food Type According to Region
CITY Adagutta Addagutta Addanki Adilabad Adilabad, Telan... Agadugutta Agiripalli
Food Type Group
Both
Non-Vegetarian
Vegetarian
Customer Count (Structured)



FAVOURITE FOOD ITEM	Total Reviews
Veg Biryani	73
Pizza	195
Other	82
Idli	83
Ice cream	114
Fried Chicken	76
Dosa	115
Chicken Biryani	203
Burger	128
Total	1663


SENTIMENT ANALYSIS
75K
Total Reviewers
36K
Positive Reviews
8808
Negative Reviews
Total Reviewers by Sentiment_Label
8.81K (11.8%)
35.61K (47....) Sentiment_L...
30.25K (40.51%)
Positive
Neutral
Negative
Avg Sentiment Score by Food Type
Food Type Group
Non-Vegetarian
Both
Vegetarian
0.00
0.05
0.10
Avg Sentiment
Most commented Text
Yummy
Very nice
Deli...
Yu...
Nice
Ex...
W...
Excellent ...
Good...
0
To...
S...
The food ...
Fine
Very tasty...
Average ...
Too much...
The food ...
The food ...
Presentat...
Tasty

CONCLUSION
Strong Customer Base with Positive Brand Perception
• The platform has 3,559 total customers and 75K total reviews, indicating strong engagement.
• Nearly 48% of reviews are positive, while negative reviews remain relatively low (~12%).
• The average rating is around 4, reflecting high overall customer satisfaction.
Food Preference Insights
• Biryani is the most loved food item, followed by Chicken Biryani and Burgers.
• Medium-calorie food (150–250 kcal range) is most preferred.
• Non-vegetarian food shows slightly higher sentiment scores compared to vegetarian.
Hyderabad Leads Market Performance
• Hyderabad stands out with the highest customer count.
• Vijayawada and Delhi follow but at a significantly lower scale.
• Regional ratings are consistently strong across cities, showing stable service quality.
Sentiment & Loyalty Patterns
• Loyal customers show strong positive sentiment.
• Critic and unhappy segments show negative sentiment but represent a smaller share.
• Sentiment scores remain moderately positive overall (~0.23 average).
Young Customers Drive Engagement
• The 18–25 age group dominates (≈78%) the customer base.
• This segment also generates the highest feedback volume.
• The platform's primary audience is clearly young adults.
Ratings Trend Shows Stability
• Daily average ratings remain consistently around 4+, with minor fluctuations.
• No significant performance drops observed, indicating stable service delivery.
Final Business Insight
FoodTrends demonstrates strong customer satisfaction, high engagement from young customers, and consistent regional performance, with Hyderabad as the key revenue-driving market.
To grow further, focus on:
• Retaining young customers (18–25 segment)
• Expanding high-performing food categories (Biryani & Non-Veg)
• Addressing feedback from critic segments
• Replicating Hyderabad's success strategy in other cities

7. Challenges Face

During the implementation of the project “Food Trends: Understanding Customer Preferences in FB ” several technical, analytical, and integration-related challenges were encountered and effectively resolved.

Unstructured Data Cleaning

Customer feedback collected from surveys, online reviews, and social media platforms contained inconsistencies such as spelling errors, abbreviations, emojis, and incomplete sentences. Preparing this unstructured data for sentiment analysis required extensive preprocessing. Text normalization, duplicate removal, handling of missing values, and formatting corrections were performed to ensure reliable analysis and accurate dashboard outputs.

Data Integration from Multiple Sources

The project involved combining structured survey data with unstructured review data. Each dataset had different column names, formats, and identifiers. Maintaining consistency across product IDs, customer IDs, and time dimensions was challenging. This issue was resolved by creating a standardized data schema and unified relational model within Microsoft Power BI to ensure seamless cross-filtering and analysis.

Sentiment Analysis Accuracy

Classifying feedback into positive, neutral, and negative categories presented challenges, especially when reviews contained mixed opinions or context-based meanings. Improving classification accuracy required refining preprocessing techniques and adjusting scoring logic to enhance reliability in the sentiment analysis pipeline.

Customer Segmentation Logic

Developing meaningful behavioral personas such as Loyal Customers, Critics, and Occasional Buyers required careful selection of parameters such as feedback frequency, sentiment score, and product preference patterns. Ensuring that the segmentation logic accurately reflected real-world customer behavior required multiple testing iterations and validation.

8. Learnings & Skills Acquired

The project offered strong exposure to Sentiment analysis, dashboard engineering, and user-centered design. Key learnings include:

- **Sentiment Analysis Implementation:** Gained hands-on experience in classifying feedback into positive, neutral, and negative categories and refining sentiment scoring logic to improve accuracy.
- **Aspect Extraction & Keyword Analysis:** Developed the ability to extract product-specific aspects such as taste, price, packaging, and quality from reviews and identify frequently occurring keywords for trend detection.
- **Dashboard Development using Microsoft Power BI:** Built interactive dashboards with sentiment trends, product comparisons, customer segmentation views, and strategic insights using DAX measures and data modeling techniques.

- **Data Modeling & Integration:** Learned to design a unified data schema integrating multiple data sources and create relationships between Customer, Product, Feedback, and Time dimensions.
- **Data Visualization & Storytelling:** Improved skills in presenting insights using bar charts, pie charts, line graphs, KPI cards, and trend visuals to effectively communicate findings to stakeholders.
- **Analytical Thinking & Problem Solving:** Enhanced ability to interpret data patterns, identify business problems, and generate actionable recommendations.
- **Teamwork & Project Management:** Strengthened collaboration, milestone tracking, documentation, and presentation skills throughout the internship.

9.Testimonials from team

The project Food Trends Understanding Customer Preferences in FB successfully demonstrated how customer feedback can be transformed into actionable business insights using business intelligence tools. By integrating sentiment analysis, product performance comparison, and customer segmentation into a unified dashboard, the project delivered a scalable solution for monitoring customer trends in the FB industry. This internship experience significantly contributed to our practical knowledge and prepared us for future roles in Business Intelligence and Data Analytics.

10.Conclusion

The project Food Trends Understanding Customer Preferences in FB Using Power BI Dashboard successfully demonstrated how customer feedback can be transformed into actionable business insights using AI and business intelligence tools. By integrating sentiment analysis, product performance comparison, and customer segmentation into a unified dashboard, the project delivered a scalable solution for monitoring customer trends in the FB industry. This internship experience significantly contributed to our practical knowledge and prepared us for future roles in Business Intelligence and Data Analytics.

11.Acknowledgements

We sincerely thank Infosys and the Infosys Springboard platform for providing this opportunity to gain industry-relevant experience. We are grateful to our mentors and coordinators for their continuous support and guidance throughout the internship journey.